Homework 8 - Engineering management

1. **Negative Pattern: Ignore Human Issues**

Description

The “Ignore Human Issues” negative pattern happens when developers focus only on technical functionality and ignore the needs and behavior of users such as pharmacists, doctors and patients.

How It Could Appear in Our Project

In the Cloud-Based Pharmacy Management System, this pattern could appear if:

- the interface is too complicated,
- pharmacists need too many steps to process prescriptions,
- doctors cannot easily manage patient records,
- users are not trained to use the system properly,
- the system sends too many unnecessary alerts.

Problems It Causes

Ignoring human issues could lead to:

- slower work,
- medication mistakes,
- frustrated users,
- reduced trust in the system.

How We Would Avoid It

To avoid this pattern, our project would:

- use a simple and user-friendly interface,
- provide role-based access for doctors, pharmacists and admins,
- reduce unnecessary steps,
- provide user training and documentation,

- create clear and important alerts only.

Conclusion

Human factors are very important in healthcare systems. Our project focuses not only on technical functionality but also on usability, safety, and user experience.

2. Negative Pattern: Last-Minute Development

Description

The "Last-Minute Development" negative pattern happens when teams delay important tasks until close to the deadline instead of working consistently throughout the project.

How It Could Appear in Our Project

In the Cloud-Based Pharmacy Management System, this pattern could appear if:

- important features are implemented too late,
- testing is postponed until the final days,
- documentation is written only before submission,
- team members delay their assigned tasks,
- bugs are discovered too late to fix properly.

Problems It Causes

Last-minute development could lead to:

- increased stress in the team,
- lower quality work,
- unfinished features,
- more software bugs,
- poor project organization.

How We Would Avoid It

To avoid this pattern, our project would:

- divide the project into smaller phases,
- set weekly goals and milestones,
- regularly monitor progress through GitHub,
- hold weekly team meetings,
- test features continuously during development.

Conclusion

Good time management is very important for successful project development. Our project focuses on continuous progress, regular communication, and organized teamwork to avoid delays and improve overall quality.

3. Positive Pattern: Continuous Integration and Testing

Description

The “Continuous Integration and Testing” positive pattern happens when developers regularly integrate their work, test features and detect problems early during development.

How It Could Appear in Our Project

In the Cloud-Based Pharmacy Management System, this pattern would be implemented by:

- regularly uploading code changes to GitHub
- testing new features after every update
- checking if prescription management, inventory tracking and patient records work correctly
- reviewing team members’ code before merging
- fixing bugs immediately when they are discovered

Benefits It Provides

Using continuous integration and testing would help:

- improve software quality
- reduce major bugs
- make teamwork more organized
- detect problems early
- increase system reliability and safety

How We Would Implement It

To implement this positive pattern, our project would:

- use GitHub for version control and collaboration
- create separate branches for new features
- test modules regularly during development
- perform code reviews before accepting changes
- maintain weekly progress checks and testing sessions

Conclusion

Continuous integration and testing are very important in healthcare software systems because they improve reliability, reduce errors and support stable system development. Our project would use regular testing, collaboration and early bug detection to maintain high software quality and ensure safe pharmacy management operations.

4. Positive Pattern: Visibility of System Status

Description

The “Visibility of System Status” positive pattern appears when the system clearly informs users what is happening, whether an action was successful and whether the displayed data is updated and reliable.

How It Could Appear in Our Project

In the Cloud-Based Pharmacy Management System, this pattern would be implemented by:

- showing a success message when a doctor adds a new patient
- showing an error message when a unique ID is invalid or already exists
- displaying whether prescription data is synced or pending
- confirming when a prescription is marked as completed

- confirming that stock was updated after dispensing medication

Benefits It Provides

Using visibility of system status would help:

- increase user trust in the system
- reduce confusion during important tasks
- prevent duplicate patient records
- reduce the risk of outdated prescriptions

How We Would Implement It

To implement this positive pattern, our project would:

- use confirmation messages after important actions
- use simple status labels such as “Synced”, “Pending update”, “Completed” and “Stock update”
- show the last update time and the person who made the change
- make important safety information such as allergies and current medicines clearly visible
- test these messages during usability testing to make sure users understand them

Conclusion

Visibility of system status is a crucial part in healthcare systems because users must know that the information is correct, updated, and safely saved. Our project would use clear messages, visible status labels, and update information to help doctors and pharmacists work faster, avoid mistakes, and trust the system more.